

Shuaikang Wang

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EDUCATION

Peking University

Beijing, China

M.Eng. in Mechanical Engineering (Rank: 1/133)

Sep. 2023 – Jul. 2026

Advisor: Prof. Meng Guo

Courses: Advanced Robotics, Reinforcement Learning, Distributed Optimization, Analytical Dynamics

Peking University

Beijing, China

B.Eng. in Robotics Engineering (GPA: 3.6/4.0)

Sep. 2019 – Jul. 2023

Advisor: Prof. Meng Guo

Courses: Machine Learning, Introduction to Intelligent Robots, Robot Perception and Control

The University of British Columbia

Vancouver, Canada

Exchange Student in the Faculty of Applied Science (GPA: 4.0/4.0)

Aug. 2021 – Dec. 2021

Courses: Introduction to Probability, Signals and Systems, Mechanics I, Applied Electronics

RESEARCH INTERESTS

My research interests lie at the intersection of **decision-making**, **learning**, and **control** in robotic systems. I am particularly focused on improving the **efficiency**, **safety**, and **flexibility** of robots in **complex tasks**.

PUBLICATIONS

[6] Intention-aware Monitoring of Dynamic Targets with Bounded Uncertainty and Minimum Fleet

Shuaikang Wang, Jianyang Xue, Yiannis Kantaros and Meng Guo

Under Review

[5] LOMORO: Long-term Monitoring of Dynamic Targets with Minimum Robotic Fleet under Resource Constraints

Mingke Lu, Shuaikang Wang, and Meng Guo

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. [PDF]

[4] Multi-UAV deployment in obstacle-cluttered environments with LOS connectivity

Yuda Chen, Shuaikang Wang, Jie Li, Meng Guo

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. [PDF]

[3] Customize Harmonic Potential Fields via Hybrid Optimization over Homotopic Paths

Shuaikang Wang, Tiecheng Guo, and Meng Guo

IEEE Robotics and Automation Letters (RA-L), 2025. [PDF] [Project]

Selected for Presentation at ICRA2026

[2] Hybrid and oriented harmonic potentials for safe task execution in unknown environment

Shuaikang Wang and Meng Guo

Automatica, 2025. [PDF] [Project]

[1] Uncertainty-bounded Active Monitoring of Unknown Dynamic Targets in Road-networks with Minimum Fleet

Shuaikang Wang, Yiannis Kantaros and Meng Guo

IEEE International Conference on Robotics and Automation (ICRA), 2024. [PDF]

Best Paper Award on Multi-Robot Systems Finalist

RESEARCH EXPERIENCE

Dexterous Grasping Synthesis via Contact-preserving Differentiable Optimization

Mar. 2025 – Sep. 2025

Research Intern, National University of Singapore

Advisor: Prof. Xingyu Liu

- Formulated a contact-preserving differentiable optimization pipeline that jointly refines the 6-DoF poses of all hand links and candidate contact points under joint-limit and non-penetration constraints.
- Derived closed-form analytic gradients for all kinematic and contact terms to ensure fast convergence.
- Implemented and benchmarked the method in *IsaacLab* across varied objects and dexterous-hand morphologies, then performed sim-to-real transfer via system identification and contact calibration.

Safe Robot Navigation in Complex Environments via Navigation Functions

Jun. 2022 – Present

Research Assistant, Peking University

Advisor: Prof. Meng Guo

- Proposed an online method to construct Navigation Functions by iteratively building diffeomorphic transformations as the environment updates, ensuring global convergence and safety from high-level task planning to low-level motion control.
- Designed a hybrid optimization scheme that jointly selects homotopy classes (discrete obstacle abstractions) and tunes continuous field parameters, yielding topology-aware, safety-preserving trajectories with certified convergence.
- Employed Flow Matching to learn diffeomorphic maps with physics-informed boundary constraints, enabling neural accelerated navigation while retaining global guarantees.

Intention-aware Monitoring of Dynamic Targets with Multi-Robot Teams

Jun. 2023 – Present

Research Assistant, Peking University

Advisor: Prof. Meng Guo and Prof. Yiannis Kantaros

- Formulated a distributed framework that couples LTL-based task specifications with Bayesian online intent inference and intent-aware Kalman prediction, enabling joint reasoning over task allocation and motion planning for moving targets.
- Designed a game-theoretic allocation scheme that attains local Nash equilibria and integrates event-triggered MPC for closed-loop execution, supporting real-time multi-robot coordination and persistent target monitoring.
- Developed a resource-constrained tasking method that minimizes team size under energy limits; validated through large-scale simulations and hardware experiments, demonstrating reduced robot usage while meeting monitoring requirements.

AWARDS & HONORS

National Scholarship (**Highest Honor for Students in China**), Ministry of Education, China

Oct. 2025

Merit Student, Peking University

Oct. 2025

Academic Innovation Award, Peking University

Oct. 2025

Academic Rising Star, College of Engineering, Peking University

May 2025

Best Paper Award on Multi-Robots Systems Finalist, IEEE ICRA

May 2024

Dean's Scholarship, College of Engineering, Peking University

Otc. 2023

(NAE) Grand Challenges Scholar Award, National Academy of Engineering, United States

Jun. 2023

Outstanding Overseas Exchange Scholarship, Peking University

Mar. 2022

Academic Excellence Award, Peking University

Oct. 2020

ACADEMIC SERVICE

Reviewer: ICRA, IROS, RA-L, CDC

Teaching Assistant: Ordinary Differential Equation (ODE), Peking University, 2024

SKILLS

Programming: Python, C, C++, MATLAB, $\text{\LaTeX}$

Frameworks: PyTorch, Anaconda, Git, ROS, IsaacLab, Mujoco, Pybullet, ManiSkill

Robots: Agile Limo, DJI Tello, Shadow Hand, Leap Hand, Unitree Go1